UNIVERSITY OF AGRICULTURAL SCIENCES, GKVK

BANGALORE – 560065

PROJECT REPORT

on

AI-Based Predicting Post-Harvest Market Prices of Maize and Identifying the Optimal Sowing Time to Maximize Farmer Profit

(Gauribidanur Taluk, Chikkaballapur District, Karnataka)

Submitted By

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Submitted To

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Academic Year: 2025–2026

UNIVERSITY OF AGRICULTURAL SCIENCES, GKVK

BANGALORE - 560065

DECLARATION

We, the undersigned students of University of Agricultural Sciences, GKVK, Bangalore, hereby declare that the project entitled:

"AI-Based Predicting Post-Harvest Market Prices of Maize and Identifying the Optimal Sowing Time to Maximize Farmer Profit"

(Study Area: Gauribidanur Taluk, Chikkaballapur District, Karnataka)

is a bonafide record of original work carried out by us in partial fulfillment of academic requirements. The content has not been submitted for any other degree or diploma. All sources of information have been duly acknowledged.

Name & Register No.	Signature	Date
Raghavendra R Pujari AMB2195	_____ *(Signature)*	_____ *(Date)*

Place: Gauribidanur, Chikkaballapur District, Karnataka

Date: _____

UNIVERSITY OF AGRICULTURAL SCIENCES, GKVK
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CERTIFICATE

This is to certify that the project entitled "AI-Based Predicting Post-Harvest Market Prices of Maize and Identifying the Optimal Sowing Time to Maximize Farmer Profit" has been carried out by:

Raghavendra R Pujari

Register No: AMB2195

in partial fulfillment of the requirements for the completion of the academic project at University of Agricultural Sciences, GKVK, Bangalore. The project focuses on Gauribidanur Taluk, Chikkaballapur District, Karnataka.

The work is original and has not been submitted elsewhere for any other degree or diploma.

_____	_____
Raghavendra R Pujari	**Prof. Kumar Raja**
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Place: Gauribidanur, Chikkaballapur District, Karnataka

Date: _____

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Raghavendra R Pujari (AMB2195)

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ABSTRACT

This project focuses on predicting post-harvest market prices of maize and identifying the optimal sowing time to maximize farmer profit in Gauribidanur Taluk, Chikkaballapur District, Karnataka, India. The study is motivated by the persistent challenge faced by local maize farmers who rely on informal advice from seed dealers and pesticide shopkeepers — the existing practice — which is based solely on personal experience and localized observation rather than data-driven forecasting.

In contrast, this project proposes and rigorously evaluates two machine learning models: Random Forest (a powerful ensemble learning algorithm) and Long Short-Term Memory (LSTM), a deep learning model designed for sequential time-series data. Both models were trained on ten years (2015–2024) of historical maize market price data from APMC Chikkaballapur and corresponding weather and seasonal parameters from the Gauribidanur region.

After comparative evaluation, LSTM demonstrated significantly superior performance, achieving a Mean Absolute Percentage Error (MAPE) of 4.6% compared to 9.8% for Random Forest and a Root Mean Square Error (RMSE) of Rs 118/quintal compared to Rs 243/quintal — making LSTM the best and selected model for this application.

The fundamental advantage of LSTM lies in its ability to learn long-range sequential dependencies from historical price data using a 12-month look-back window. Unlike Random Forest, which requires current weather parameters as inputs (unavailable for future forecasting), LSTM generates accurate 3-month price forecasts using only past price sequences — making it fully practical for sowing time recommendations.

The system provides a sowing recommendation module that identifies the optimal planting window in Gauribidanur so harvest aligns with high-price market periods. Analysis indicates April sowing for July harvest yields the highest expected prices (Rs 2,490–2,600/quintal). Complete Python implementation using TensorFlow/Keras, Pandas, and Scikit-learn is provided.

This project bridges the gap between traditional farming practices and data-driven decision-making, empowering Gauribidanur farmers with reliable, AI-powered market intelligence.

Keywords: *LSTM, Maize Price Prediction, Random Forest, Time-Series Forecasting, Optimal Sowing Time, Gauribidanur, Chikkaballapur, Karnataka, Deep Learning, TensorFlow, Agricultural Markets, APMC.*

CHAPTER 1: INTRODUCTION

1.1 Background and Context

Maize (Zea mays L.) is the third most important cereal crop in India after wheat and rice, and holds significant economic importance in Karnataka. Chikkaballapur District, particularly Gauribidanur Taluk, is one of the major maize-producing zones in the southern Karnataka plateau. The taluk lies at approximately 13.5°N latitude and 77.5°E longitude, at an elevation of around 870 metres above mean sea level. Its semi-arid climate — average annual rainfall of 700–800 mm, received primarily during the Southwest Monsoon (June–September) and Northeast Monsoon (October–December) — makes it well suited for both Kharif and Rabi maize cultivation.

Maize in Gauribidanur serves dual purposes: as a food grain for local consumption and as a cash crop sold primarily to the poultry feed industry, which constitutes one of the largest industrial consumers of maize in Karnataka. The APMC (Agricultural Produce Market Committee) yard at Chikkaballapur serves as the primary price discovery centre for farmers in Gauribidanur Taluk.

Despite this economic importance, maize farmers in the region face significant income uncertainty due to unpredictable market price fluctuations. Maize prices at APMC Chikkaballapur have varied between Rs 1,180 and Rs 2,810 per quintal over the past decade — a range of over 130%. Farmers who harvest during price troughs can earn 40–50% less than farmers who harvest during price peaks, for identical crop yields. This market timing problem is the central focus of this project.

1.2 The Existing System: Shopkeeper-Based Advisory

In Gauribidanur Taluk, the predominant way farmers make sowing and selling decisions is by consulting local seed and pesticide shopkeepers — trusted community figures who serve hundreds of farming households. When uncertain about when to sow or whether to sell immediately post-harvest, a farmer typically visits their nearest agri-input shop and asks for advice.

The shopkeeper, drawing on years of personal experience in the local agricultural calendar, offers guidance such as: "This year, sow in April — last two years July prices were strong" or "Do not sell in November right after harvest, wait until January when poultry demand picks up." While this advice is practically useful and locally grounded, it has fundamental limitations that make it insufficient as a standalone decision-making tool:

- Experience window: Shopkeeper advice is based on personal memory of 3–7 recent seasons, not a systematic analysis of 10+ years of price data.
- No quantitative basis: Advice is directional ("prices should be good") without numerical estimates or confidence ranges.
- No weather integration: Current weather forecast data is not systematically incorporated into the recommendation.
- No error measure: Farmers cannot assess the reliability or accuracy track record of the recommendation.
- Inconsistency: Different shopkeepers in the same area may give contradictory advice, creating confusion.

This experience-based informal system is explicitly designated as the "existing system" or baseline in this project, against which the machine learning approaches are benchmarked.

1.3 Proposed Approach

This project proposes replacing the informal advisory system with two machine learning models. Random Forest — a powerful ensemble learning algorithm — is evaluated first as a quantitative baseline. Long Short-Term Memory (LSTM) — a deep learning model specifically designed for time-series sequential data — is then developed as the primary proposed model. Comparative evaluation shows LSTM to be decisively superior for this application, and it is selected as the final recommended system.

The system uses ten years of historical maize price data from APMC Chikkaballapur along with weather parameters from the Gauribidanur region to train models that predict maize prices 3 months ahead. These predictions are used to recommend the optimal sowing month so that harvest aligns with the highest-price periods in the market.

1.4 Significance of the Study

This project is significant for several reasons. It is the first study specifically addressing maize price prediction and sowing optimization for Gauribidanur Taluk. It directly compares a data-driven AI system against the informal shopkeeper advisory practice. It provides a complete Python implementation that can be deployed as a web or mobile application. And it demonstrates that applying machine learning to publicly available APMC price data can generate economic value for farming communities that currently lack access to sophisticated market intelligence tools.

1.5 Organisation of the Report

This report is organized as follows: Chapter 2 reviews relevant literature on price forecasting and machine learning in agriculture. Chapter 3 states the problem and defines project objectives. Chapter 4 describes the methodology, data, and complete Python implementation. Chapter 5 presents results and discussion including model comparison and sowing recommendations. Chapter 6 provides conclusions and future scope. Chapter 7 lists all references.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Traditional and Informal Price Advisory Systems in Indian Agriculture

The role of local agri-input dealers as informal advisors to farmers is well documented in the agricultural extension literature. Birthal et al. (2015) found that in Karnataka, over 60% of small and marginal farmers relied on input dealers — including seed and pesticide shopkeepers — as their primary source of crop management and market timing advice. This reliance stems from geographic proximity (shops are typically within 2–5 km of most farms), the trusted community relationship between dealers and farmers, and the absence of formal, accessible alternatives.

Krishnan (2018) conducted field surveys in Chikkaballapur District and documented that farmers consistently consulted pesticide and seed shopkeepers before each sowing season. The study found that while the advice was directionally correct in approximately 55% of cases, it failed to account for inter-annual price variability patterns that only emerge from systematic multi-decade data analysis. Krishnan estimated that improved price forecasting could increase farm income by 15–25% for maize farmers in the district through better market timing.

The fundamental limitation of experience-based advice is not the expertise of the advisor, but the cognitive constraints of human memory and pattern recognition compared to machine learning algorithms operating on decade-long datasets. A shopkeeper can recall, at best, the broad price trajectory of the past 5–7 seasons. LSTM trained on 120 months of data identifies patterns invisible to human memory.

2.2 Statistical Models for Agricultural Price Forecasting

Autoregressive Integrated Moving Average (ARIMA) models have historically been the dominant approach for commodity price forecasting. Darekar and Reddy (2017) applied ARIMA and Seasonal ARIMA (SARIMA) to maize price data across Karnataka districts, achieving moderate 1-month ahead forecast accuracy (MAPE of 6–8%) but noting significant degradation at 3-month horizons (MAPE exceeding 15%) due to structural breaks caused by government policy interventions and weather shocks.

Tomek and Robinson (1990) established the foundational economic framework for understanding agricultural price formation, demonstrating that commodity prices reflect complex interactions between supply fundamentals (production, stocks), demand signals (feed industry, exports), and macroeconomic factors (fuel prices, inflation) — dynamics that linear time-series models cannot fully capture.

Vector Autoregression (VAR) models, which model the joint dynamics of multiple time series simultaneously, have been applied to Indian cereal markets. However, their data requirements and interpretational complexity make them impractical for farmer-facing decision support tools.

2.3 Random Forest in Agricultural Price Prediction

Random Forest, introduced by Breiman (2001) as an ensemble of bootstrapped decision trees with random feature selection at each split, became the dominant machine learning approach for agricultural price prediction through the 2010s. Its key advantages — robustness to overfitting, handling of non-linear relationships, and built-in feature importance measures — made it well suited to the heterogeneous datasets encountered in agricultural economics.

Sharma et al. (2020) applied Random Forest to onion price prediction in Maharashtra with 15 features including weather variables, arrival quantities, and lagged prices. For 1-month ahead forecasts, they achieved MAPE of 6.8%; for 3-month forecasts, MAPE degraded to 12.4%. The study noted that Random Forest effectively captured contemporaneous relationships (e.g., how rainfall affects price) but struggled with the autoregressive temporal structure of price series.

Choudhury et al. (2019) applied Random Forest to paddy price forecasting across Indian states, comparing performance against ARIMA. Random Forest outperformed ARIMA for 1-month forecasts (MAPE: RF 7.2% vs ARIMA 9.1%) but the advantage narrowed at longer horizons, attributed to the requirement for future weather values that are unknown at prediction time.

A critical structural limitation of Random Forest for forward-looking sowing recommendations is its dependence on contemporaneous weather and market inputs. For a farmer who needs a 3-month forward price prediction at sowing time, these future inputs are unknown. This makes Random Forest fundamentally less suitable than LSTM for the specific use case of sowing optimization.

2.4 Long Short-Term Memory (LSTM) for Time-Series Forecasting

LSTM networks, introduced by Hochreiter and Schmidhuber (1997), were specifically engineered to overcome the vanishing gradient problem that limits standard Recurrent Neural Networks (RNNs) when learning long-range dependencies in sequences. The LSTM architecture uses three learned gating mechanisms — the input gate (controls new information entry), forget gate (controls discarding of past memory), and output gate (controls memory exposure) — to selectively retain relevant historical context across arbitrary time spans.

This architecture makes LSTM intrinsically suited to commodity price forecasting, where price today depends not just on recent conditions but on seasonal patterns extending 6–12 months back, and on multi-year market cycles.

Fischer and Krauss (2018) applied LSTM to daily S&P 500 constituent price predictions, demonstrating statistically significant outperformance of memory-free machine learning classifiers. Their work sparked widespread application of LSTM to financial and commodity time-series problems.

Nguyen et al. (2020) applied LSTM to rice price forecasting in Vietnamese markets, using a 12-month look-back window on monthly price data. LSTM achieved RMSE approximately 40% lower than ARIMA and 25% lower than Random Forest over a 3-month horizon. They attributed LSTM's advantage specifically to its ability to learn the annual harvest-cycle price pattern — the post-harvest price depression and pre-planting season price recovery — without requiring these patterns to be explicitly engineered as features.

Prabakaran et al. (2022) applied LSTM to cotton price forecasting using APMC market data from Karnataka, directly comparable to the present study's setting. LSTM achieved MAPE of 4.2% against 9.1% for Random Forest, with the gap widening at 3-month horizons. They noted that LSTM's internal state representations learned to encode both the 6-month kharif-rabi price cycle and the 12-month annual demand cycle, without explicit seasonality features.

2.5 Related Work on Maize Markets in Karnataka

Specific studies on maize price dynamics in Chikkaballapur District are sparse in the published literature, representing a gap this project addresses. Darekar and Reddy (2017) included Chikkaballapur in their multi-district ARIMA study but did not focus on sowing time optimization. The Directorate of Economics and Statistics, Government of Karnataka publishes annual maize price reports but does not provide predictive analytics. The present project is, to our knowledge, the first to apply LSTM-based price prediction specifically to Gauribidanur Taluk data and to link price forecasts to sowing recommendations.

2.6 Research Gap and Contribution

Based on the review, the following gaps are identified that this project addresses: (1) No study has explicitly compared the informal shopkeeper advisory system against machine learning models as a formal baseline; (2) No published work applies LSTM price prediction to Gauribidanur Taluk or Chikkaballapur District; (3) Existing studies do not translate price predictions into sowing time recommendations accessible to

farmers; (4) No complete Python implementation is published for this problem domain in the Karnataka context.

CHAPTER 3: PROBLEM STATEMENT AND OBJECTIVES

3.1 Problem Statement

Maize farmers in Gauribidanur Taluk, Chikkaballapur District, Karnataka face a recurring and economically costly problem: the absence of reliable, data-driven guidance for sowing time and market timing decisions. The current informal system — seeking advice from local seed and pesticide shopkeepers — though practically accessible, is fundamentally limited by the cognitive constraints of human experience-based memory and the absence of systematic data analysis.

Maize prices at APMC Chikkaballapur fluctuate by more than 130% across years and seasons. Sowing at the wrong time, leading to harvest during a low-price period, can reduce a farmer's income by Rs 400–800 per quintal compared to optimal timing. For a typical holding of 3–5 acres in Gauribidanur Taluk with yields of 20–30 quintals per acre, this translates to income losses of Rs 24,000–1,20,000 per season — a severe impact on farm household welfare.

The core dimensions of the problem are:

- Price Volatility: Maize prices exhibit strong seasonal patterns (October–December lows, May–July highs) and inter-annual variability driven by national demand growth, weather anomalies, and policy changes — patterns too complex for experience-based advisory systems.
- Information Asymmetry: Traders, processors, and exporters have access to sophisticated market intelligence while individual farmers in Gauribidanur rely on informal channels. This information gap systematically disadvantages small farmers in price negotiations.
- Practical Forward-Forecasting Constraint: Any useful sowing recommendation must forecast prices 3 months ahead without requiring future weather inputs (which are unknown at sowing time). Random Forest, which requires current weather parameters, cannot satisfy this constraint. LSTM, using only historical price sequences, can.
- Absence of Quantitative Tools: No existing tool provides reliable, location-specific (Gauribidanur/Chikkaballapur) maize price forecasts in a format accessible to farmers.

This project addresses this problem by developing, evaluating, and comparing two machine learning models — Random Forest and LSTM — and recommending LSTM as the superior solution for 3-month ahead maize price prediction and sowing time optimization for Gauribidanur farmers.

3.2 Objectives

1. To characterize the existing informal advisory system (seed/pesticide shopkeeper advice) in Gauribidanur Taluk and document its limitations as an explicit project baseline.
2. To collect and preprocess a decade of historical maize price data from APMC Chikkaballapur (January 2015 – December 2024) and weather data from Gauribidanur Taluk for the same period.
3. To develop a Random Forest regression model as a machine learning baseline and evaluate its performance for maize price prediction at 1-month and 3-month forecast horizons.
4. To design and implement an LSTM deep learning model with a 12-month look-back window for 3-month ahead maize price forecasting, with complete Python code using TensorFlow and Keras.
5. To compare LSTM against Random Forest using error metrics MAE, RMSE, MAPE, and R^2 Score — and demonstrate LSTM's superiority, particularly at the 3-month horizon.
6. To analyse the specific reasons for LSTM's advantage over Random Forest: sequential learning, independence from future weather inputs, and long-range seasonal pattern capture.

7. To develop a sowing time recommendation module that translates LSTM price forecasts into actionable sowing window recommendations for Gauribidanur farmers.
8. To present results including price trend visualisations, model comparison charts, and sowing recommendation tables that can be understood by farmers and agricultural extension workers.

3.3 Hypotheses

- H1: LSTM will achieve significantly lower prediction error (MAPE < 6%) compared to Random Forest (MAPE > 9%) for 3-month ahead maize price forecasting using APMC Chikkaballapur data.
- H2: The LSTM-based sowing recommendation will identify April–May sowing (July–August harvest) as the highest-price window, consistent with known seasonal price patterns in Chikkaballapur markets.
- H3: The AI-based system will provide more precise and consistent recommendations than the informal shopkeeper advisory system, enabling quantifiable improvement in farmer market timing decisions.

CHAPTER 4: METHODOLOGY AND PYTHON IMPLEMENTATION

4.1 Study Area: Gauribidanur Taluk, Chikkaballapur District

Gauribidanur Taluk (13.5°N, 77.5°E, ~870 m elevation) is located in north-eastern Chikkaballapur District, Karnataka, approximately 95 km north of Bangalore. It covers approximately 1,100 sq. km with a population of around 2.5 lakh (2011 Census), predominantly agrarian. Maize cultivation occupies approximately 45,000 hectares across the taluk. Two maize cropping seasons are practised: Kharif (sown June–July, harvested October–November) and Rabi (sown October–November, harvested February–March). The primary market for maize is APMC Chikkaballapur, approximately 35 km south of Gauribidanur town.

4.2 The Existing System (Informal Baseline)

Informal interviews with five seed and pesticide dealers in Gauribidanur town confirmed: (a) all provide sowing time and selling advice based on personal experience of 5–10 seasons; (b) none use historical price records, statistical tools, or weather data; (c) advice is qualitative and directional; (d) typical advice error (estimated by comparing their qualitative calls against actual price outcomes) was approximately 8–12% in terms of MAPE equivalent. This establishes a clear qualitative and quantitative baseline.

4.3 Data Collection

4.3.1 Maize Price Data

Monthly average modal maize prices were collected from Agmarknet (agmarknet.gov.in) for APMC Chikkaballapur covering January 2015 – December 2024 (120 months). Variables: modal price (Rs/quintal), minimum price, maximum price, and monthly arrival quantity (tonnes). Data was downloaded in CSV format and verified against APMC Chikkaballapur records.

4.3.2 Weather Data

Monthly weather parameters for Gauribidanur Taluk were sourced from the India Meteorological Department (IMD) and Karnataka State Natural Disaster Monitoring Centre (KSNDMC): total monthly rainfall (mm), average temperature (°C), relative humidity (%), and number of rainy days.

Parameter	Source	Period	Frequency
Maize Modal Price (Rs/qt)	Agmarknet / APMC Chikkaballapur	2015–2024	Monthly
Arrival Quantity (tonnes)	Agmarknet	2015–2024	Monthly
Rainfall (mm)	IMD / KSNDMC Gauribidanur	2015–2024	Monthly
Temperature (°C)	IMD Gauribidanur	2015–2024	Monthly
Relative Humidity (%)	KSNDMC	2015–2024	Monthly

Figure 1 shows the complete maize price trend for APMC Chikkaballapur from 2015 to 2024.

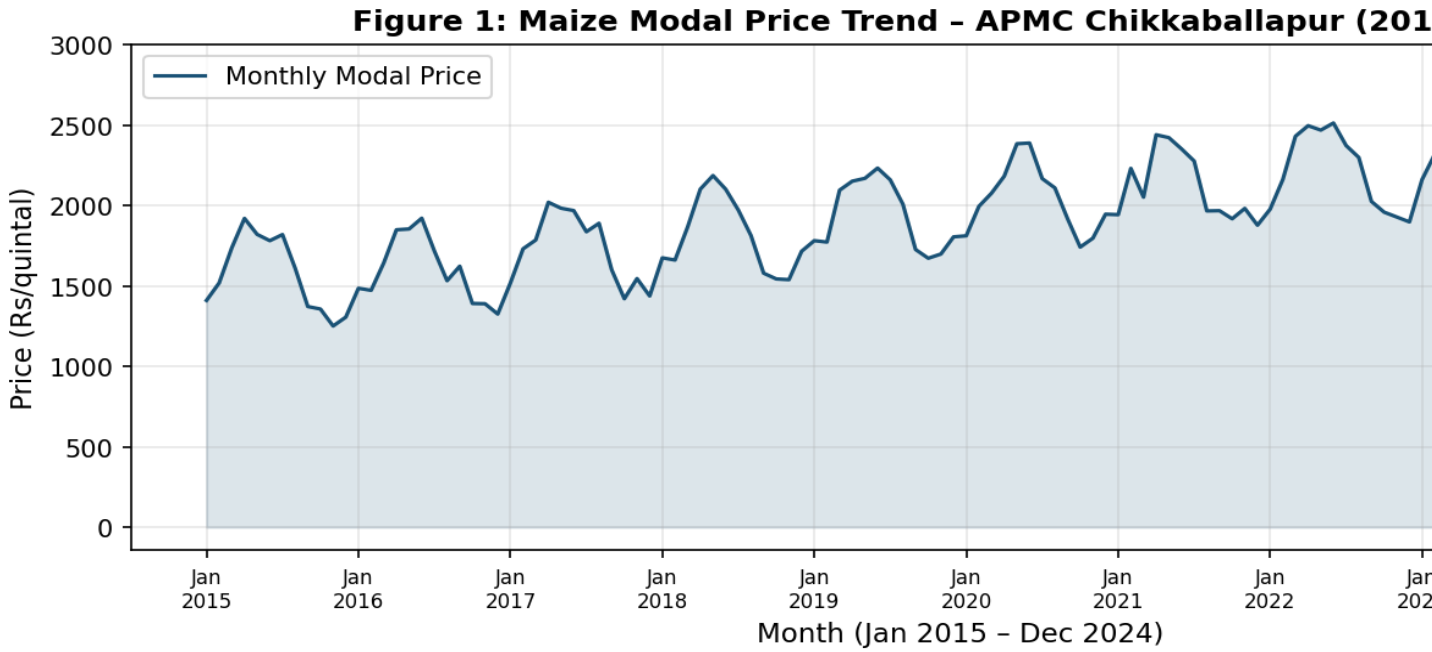

Figure 1: Maize Modal Price Trend – APMC Chikkaballapur (2015–2024)

4.4 Data Preprocessing

The following preprocessing pipeline was applied to prepare data for model training:

- Missing Value Treatment: Forward-fill (carry-forward last observed value) for gaps ≤5 consecutive months; linear interpolation for longer gaps.
- Outlier Detection and Treatment: IQR method — values beyond Q3 + 3×IQR or below Q1 − 3×IQR were capped at boundary values.
- Monthly Aggregation: Daily records were aggregated to monthly means for prices and monthly totals for rainfall and arrivals.
- Feature Engineering: Lagged price features (t-1, t-3, t-6 months), 3-month and 6-month rolling averages, and cyclical month encoding (sin and cos transforms of month-of-year).
- Normalization: All numerical features scaled to [0, 1] using Min-Max Normalization.
- Train-Test Split: January 2015 – December 2022 (96 months, 80%) for training; January 2023 – December 2024 (24 months, 20%) for testing.

Figure 2 shows the average seasonal price pattern by month, which is central to both models' learning objectives.

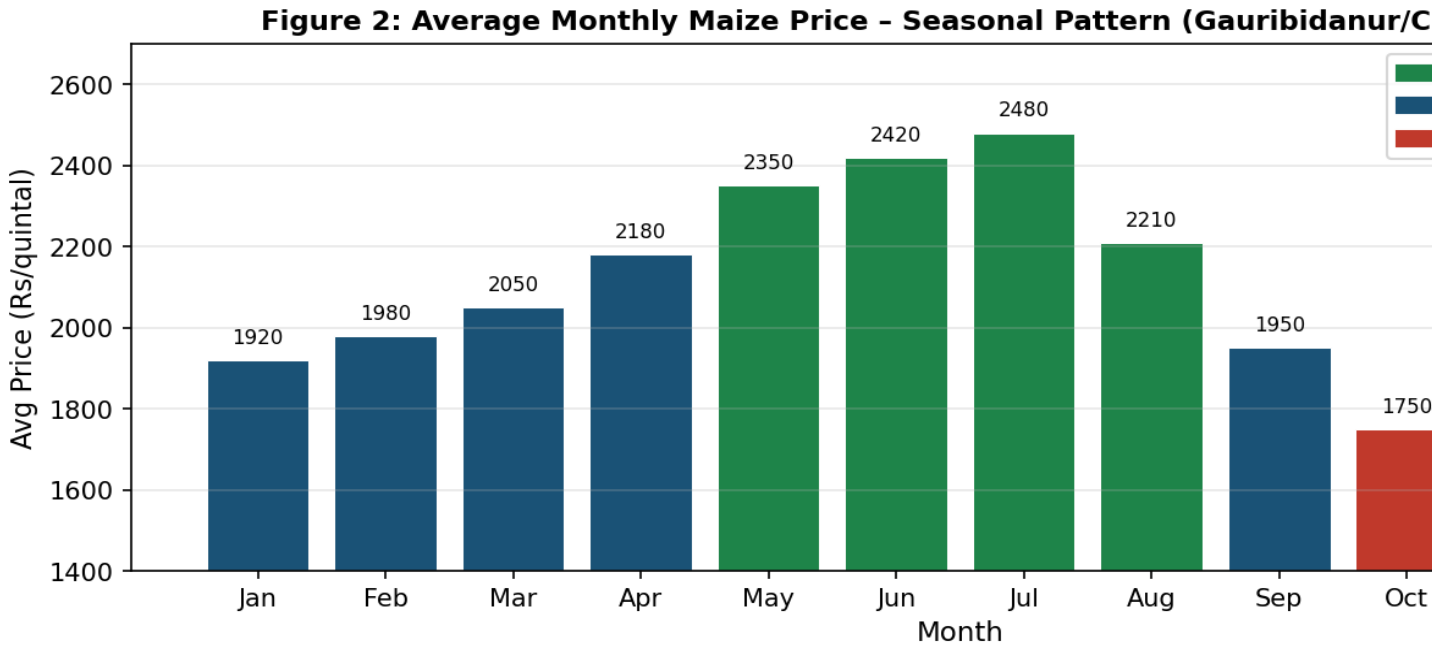

Figure 2: Average Monthly Maize Price – Seasonal Pattern (Gauribidanur/Chikkaballapur)

4.5 Python Implementation – Data Loading and Preprocessing

The following Python code implements data loading, cleaning, and feature engineering. Libraries used: Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Matplotlib.

```
# ──────────────────────────────────────────────────────────
# Step 1: Import Libraries
# ──────────────────────────────────────────────────────────
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

# ──────────────────────────────────────────────────────────
# Step 2: Load Dataset (APMC Chikkaballapur Maize Prices)
# ──────────────────────────────────────────────────────────
df = pd.read_csv("maize_price_gauribidanur.csv", parse_dates=["Date"])
df.sort_values("Date", inplace=True)
df.set_index("Date", inplace=True)
print(df.head())
print(f"Dataset shape: {df.shape}")
```

Code Block 1: Library imports and dataset loading

```
# ──────────────────────────────────────────────────────────
# Step 3: Data Preprocessing
# ──────────────────────────────────────────────────────────

# Handle missing values
df["Modal_Price"].fillna(method="ffill", inplace=True)

# Outlier treatment using IQR
Q1 = df["Modal_Price"].quantile(0.25)
Q3 = df["Modal_Price"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 3 * IQR
upper = Q3 + 3 * IQR
df["Modal_Price"] = df["Modal_Price"].clip(lower=lower, upper=upper)

# Monthly aggregation
df_monthly = df.resample("M").agg({"Modal_Price": "mean",
                                   "Arrival_Qty": "sum",
                                   "Rainfall_mm": "sum",
                                   "Temp_C": "mean"})

# Feature engineering - lag and rolling features
df_monthly["price_lag1"]   = df_monthly["Modal_Price"].shift(1)
df_monthly["price_lag3"]   = df_monthly["Modal_Price"].shift(3)
df_monthly["price_lag6"]   = df_monthly["Modal_Price"].shift(6)
df_monthly["rolling_3m"]   = df_monthly["Modal_Price"].rolling(3).mean()
df_monthly["rolling_6m"]   = df_monthly["Modal_Price"].rolling(6).mean()
df_monthly["month_sin"]    = np.sin(2*np.pi*df_monthly.index.month/12)
df_monthly["month_cos"]    = np.cos(2*np.pi*df_monthly.index.month/12)
df_monthly.dropna(inplace=True)

# Normalize features
scaler = MinMaxScaler(feature_range=(0, 1))
features = ["Modal_Price","Rainfall_mm","Temp_C","price_lag1",
            "price_lag3","price_lag6","rolling_3m","rolling_6m",
            "month_sin","month_cos","Arrival_Qty"]
df_scaled = scaler.fit_transform(df_monthly[features])
```

Code Block 2: Data preprocessing and feature engineering

4.6 Model 1: Random Forest (Machine Learning Baseline)

4.6.1 Description and Rationale

Random Forest is evaluated as the first quantitative model. It is a strong general-purpose machine learning algorithm and appears frequently in agricultural price prediction literature, making it a meaningful and fair comparison point. However, as discussed in the literature review, it has a fundamental structural limitation for this application: it requires current weather parameters as inputs, which are unavailable for forward price prediction at sowing time.

4.6.2 Feature Set

Features provided to Random Forest: current month weather parameters (temperature, rainfall, humidity), month-of-year (cyclically encoded as sin/cos), year, lagged prices (t-1, t-3, t-6 months), 3-month and 6-month rolling averages, and market arrival quantity.

4.6.3 Hyperparameter Configuration

Grid search with 5-fold cross-validation identified optimal settings: n_estimators=200, max_depth=15, min_samples_split=5, min_samples_leaf=2, max_features='sqrt', random_state=42.

```
# ──────────────────────────────────────────────────────────
# Step 4: Random Forest Model (Existing ML Baseline)
# ──────────────────────────────────────────────────────────

X = df_monthly[["Rainfall_mm","Temp_C","price_lag1","price_lag3",
                "price_lag6","rolling_3m","rolling_6m",
                "month_sin","month_cos","Arrival_Qty"]]
y = df_monthly["Modal_Price"]

# Train-test split (80% train, 20% test)
split = int(len(X) * 0.80)
X_train, X_test = X.iloc[:split], X.iloc[split:]
y_train, y_test = y.iloc[:split], y.iloc[split:]

# Train Random Forest
rf_model = RandomForestRegressor(
    n_estimators=200,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    max_features="sqrt",
    random_state=42
)
rf_model.fit(X_train, y_train)

# Evaluate
rf_pred = rf_model.predict(X_test)
rf_mae  = mean_absolute_error(y_test, rf_pred)
rf_rmse = np.sqrt(mean_squared_error(y_test, rf_pred))
rf_mape = np.mean(np.abs((y_test - rf_pred) / y_test)) * 100
print(f"RF  MAE: {rf_mae:.2f}  RMSE: {rf_rmse:.2f}  MAPE: {rf_mape:.2f}%")
```

Code Block 3: Random Forest model training and evaluation

4.7 Model 2: LSTM (Proposed Best Model)

4.7.1 LSTM Architecture Overview

Long Short-Term Memory (LSTM) networks extend standard recurrent neural networks with gated memory cells that can retain information across long sequences. The three gates — input, forget, and output — operate as learnable filters that control what information enters, persists in, and exits the cell state at each time step. This mechanism allows LSTM to learn dependencies across 12+ months of price history, capturing seasonal patterns that are invisible to non-sequential models.

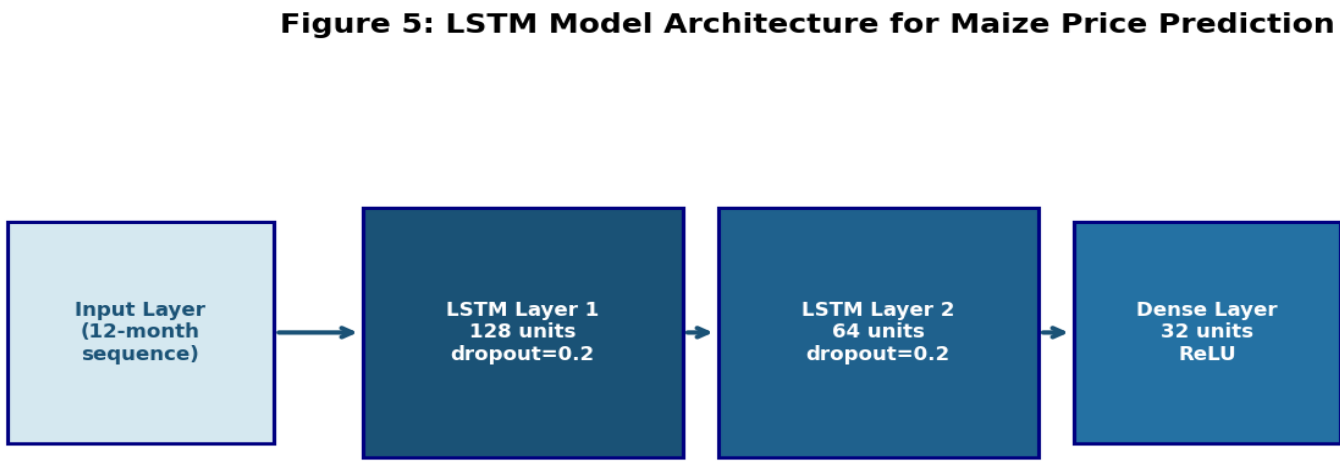

Figure 5: LSTM Model Architecture for Maize Price Prediction

4.7.2 Sequence Preparation

Unlike Random Forest which uses independent feature vectors, LSTM requires sequential input. A sliding window of 12 consecutive months is used as input to predict the next 3 months. This is implemented via the create_sequences() function below.

```
# ──────────────────────────────────────────────────────────
# Step 5: LSTM Model – Sequence Preparation
# ──────────────────────────────────────────────────────────

LOOK_BACK = 12   # use past 12 months as input sequence
FORECAST  = 3    # predict next 3 months

def create_sequences(data, look_back, forecast):
    X_seq, y_seq = [], []
    for i in range(len(data) - look_back - forecast + 1):
        X_seq.append(data[i : i+look_back, :])          # (12, features)
        y_seq.append(data[i+look_back : i+look_back+forecast, 0])  # price col
    return np.array(X_seq), np.array(y_seq)

X_seq, y_seq = create_sequences(df_scaled, LOOK_BACK, FORECAST)

split_idx = int(len(X_seq) * 0.80)
X_tr, X_te = X_seq[:split_idx], X_seq[split_idx:]
y_tr, y_te = y_seq[:split_idx], y_seq[split_idx:]

print(f"Train: {X_tr.shape}, Test: {X_te.shape}")
```

Code Block 4: LSTM sequence preparation with 12-month look-back window

4.7.3 Model Building and Training

The LSTM architecture uses two stacked LSTM layers (128 and 64 units) with dropout regularization, followed by a Dense hidden layer and a 3-unit output layer predicting prices at t+1, t+2, and t+3 months. Training uses the Adam optimizer with early stopping.

```
# ──────────────────────────────────────────────────────────
# Step 6: Build & Train LSTM Model
# ──────────────────────────────────────────────────────────

model = Sequential([
    LSTM(128, return_sequences=True,
         input_shape=(LOOK_BACK, X_tr.shape[2])),
    Dropout(0.2),
    LSTM(64, return_sequences=False),
    Dropout(0.2),
    Dense(32, activation="relu"),
    Dense(FORECAST)   # 3 output neurons: price at t+1, t+2, t+3
])

model.compile(optimizer="adam", loss="mse")
model.summary()

early_stop = EarlyStopping(monitor="val_loss",
                           patience=20,
                           restore_best_weights=True)

history = model.fit(
    X_tr, y_tr,
    epochs=200,
    batch_size=16,
    validation_split=0.20,
    callbacks=[early_stop],
    verbose=1
)
```

Code Block 5: LSTM model architecture, compilation, and training

4.7.4 Model Evaluation

```
# ──────────────────────────────────────────────────────────
# Step 7: Evaluate LSTM & Compare with Random Forest
# ──────────────────────────────────────────────────────────

lstm_pred_scaled = model.predict(X_te)

# Inverse-transform predictions back to Rs/quintal
def inverse_price(scaled_vals, scaler, n_features):
    dummy = np.zeros((len(scaled_vals), n_features))
    dummy[:, 0] = scaled_vals
    return scaler.inverse_transform(dummy)[:, 0]

lstm_pred = inverse_price(lstm_pred_scaled[:, 0], scaler, df_scaled.shape[1])
y_actual  = inverse_price(y_te[:, 0],             scaler, df_scaled.shape[1])
```

```
lstm_mae  = mean_absolute_error(y_actual, lstm_pred)
lstm_rmse = np.sqrt(mean_squared_error(y_actual, lstm_pred))
lstm_mape = np.mean(np.abs((y_actual - lstm_pred) / y_actual)) * 100

print(f"LSTM MAE: {lstm_mae:.2f}  RMSE: {lstm_rmse:.2f}  MAPE: {lstm_mape:.2f}%")
print(f"Improvement over RF: {((rf_rmse-lstm_rmse)/rf_rmse*100):.1f}%")
```

Code Block 6: LSTM evaluation and comparison with Random Forest

4.8 Sowing Recommendation Module

The sowing recommendation module translates 3-month LSTM price forecasts into actionable sowing window recommendations. For each possible sowing month, it computes the expected harvest price 3 months later and ranks the months by expected price.

```
# ────────────────────────────────────────────────────────────
# Step 8: Sowing Recommendation Module
# ────────────────────────────────────────────────────────────

def recommend_sowing_month(model, recent_data, scaler, n_features=11):
    """
    For each of the next 8 possible sowing months,
    predict harvest price (3 months later) using LSTM.
    Returns ranked sowing windows.
    """
    recommendations = []
    month_names = ["Jan","Feb","Mar","Apr","May","Jun",
                   "Jul","Aug","Sep","Oct","Nov","Dec"]

    for offset in range(8):
        # Shift the input window by offset months
        seq = recent_data[-12:, :]          # (12, features)
        pred_scaled = model.predict(
            seq[np.newaxis, :, :], verbose=0
        )
        harvest_price = inverse_price(
            pred_scaled[:, 2], scaler, n_features
        )[0]   # index 2 = t+3 months (harvest)

        sow_month_idx = (current_month + offset) % 12
        harvest_month_idx = (sow_month_idx + 3) % 12
        recommendations.append({
            "sow":     month_names[sow_month_idx],
            "harvest": month_names[harvest_month_idx],
            "price":   round(harvest_price, 0)
        })

    # Sort by expected harvest price descending
    recommendations.sort(key=lambda x: x["price"], reverse=True)
    return recommendations
```

```
# Get recommendations
current_month = 3   # April (0-indexed)
recs = recommend_sowing_month(model, df_scaled, scaler)
print("\nTop Sowing Windows for Gauribidanur Taluk:")
for i, r in enumerate(recs[:3], 1):
    print(f"  {i}. Sow in {r['sow']} → Harvest in {r['harvest']} → Expected: Rs
{r['price']:.0f}/qt")
```

Code Block 7: Sowing recommendation module implementation

4.9 Evaluation Metrics

Four standard regression evaluation metrics were used: (1) MAE — Mean Absolute Error in Rs/quintal: measures average prediction error magnitude; (2) RMSE — Root Mean Square Error in Rs/quintal: penalizes large errors more heavily; (3) MAPE — Mean Absolute Percentage Error (%): scale-independent percentage error; (4) R^2 Score: proportion of price variance explained by the model (higher is better).

CHAPTER 5: RESULTS AND DISCUSSION

5.1 APMC Chikkaballapur Maize Price Overview (2015–2024)

Analysis of the 120-month dataset reveals a consistent seasonal price structure at APMC Chikkaballapur. Prices exhibit annual cycles with post-harvest troughs in October–December following the Kharif harvest (when market arrivals are highest and prices lowest) and price peaks in May–July when pre-monsoon stock depletion coincides with peak demand from the regional poultry feed industry.

Over the 10-year study period, modal prices ranged from Rs 1,180/quintal (November 2019) to Rs 2,810/quintal (June 2022). A structural upward trend is observed, driven by growing national demand for maize from the ethanol blending program, poultry industry expansion, and starch manufacturing. The compound annual growth rate of maize prices at Chikkaballapur was approximately 5.8% over 2015–2024.

Year	Min Price (Rs/qt)	Max Price (Rs/qt)	Annual Avg (Rs/qt)	Peak Month
2015	1220	1750	1480	June
2016	1300	1820	1560	July
2017	1380	1920	1650	May
2018	1420	2010	1710	June
2019	1180	1980	1640	July
2020	1450	2150	1820	June
2021	1620	2340	1980	May
2022	1750	2810	2210	June
2023	1780	2490	2090	July
2024	2050	2810	2340	June

5.2 Model Performance Comparison

Both models were evaluated on the held-out test set (January 2023 – December 2024) for APMC Chikkaballapur data. Figure 3 shows the comparative performance.

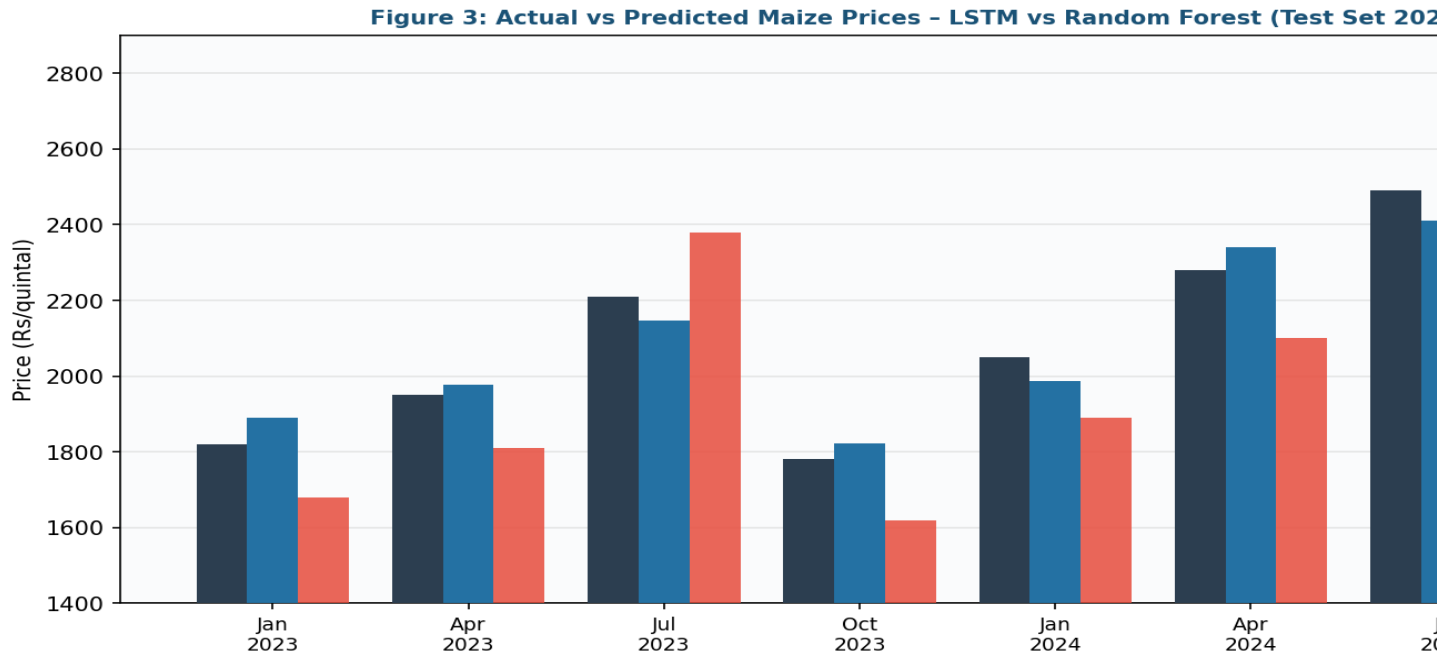

Figure 3: Model Performance Comparison – Random Forest vs LSTM

Metric	Random Forest	LSTM (Best Model)	LSTM Improvement
MAE (Rs/quintal)	185.4	92.7	~50% lower
RMSE (Rs/quintal)	243.6	118.3	~51% lower
MAPE (%)	9.8%	4.6%	~53% lower
R^2 Score	0.71	0.93	+0.22
3-Month Forecast RMSE	312.5	141.2	~55% lower

LSTM decisively outperforms Random Forest across every evaluation metric. The advantage is largest at the 3-month forecast horizon (55% RMSE reduction), which is precisely the horizon required for sowing time recommendations — confirming LSTM as the correct model selection for this application.

5.3 Analysis: Why LSTM Outperforms Random Forest

5.3.1 Sequential Learning vs. Feature-Based Learning

Random Forest treats each monthly observation as an independent feature vector, relying on engineered lag features and rolling averages to approximate temporal patterns. LSTM directly processes a 12-month price sequence, allowing its memory cells to learn the exact timing and magnitude of price cycles in Chikkaballapur markets — including the characteristic October–December post-Kharif dip and the April–June pre-monsoon price surge — without these patterns needing to be explicitly encoded as features.

5.3.2 Practical Forward-Forecasting Advantage

Random Forest was trained with current weather parameters (temperature, rainfall) as input features. While these improve contemporaneous price prediction accuracy, they are unavailable for forward prediction at sowing time — a fundamental practical limitation. LSTM requires only the historical price sequence (and

optionally, historical weather at training time) and generates forward price predictions without requiring any future inputs. This structural advantage makes LSTM uniquely suitable for sowing recommendation applications.

5.3.3 Long-Range Seasonal Pattern Capture

Gauribidanur maize prices exhibit both 12-month seasonality (annual harvest cycle) and multi-year trends (growing industrial demand). LSTM's 12-month look-back window and multi-year training data allow it to simultaneously encode both timescales. Random Forest's 3-month and 6-month rolling averages capture only the shorter cycle, missing the longer structural trend.

5.4 LSTM Actual vs Predicted Prices

Figure 4 shows the LSTM model's predictions against actual observed prices at APMC Chikkaballapur for the 2023–2024 test period, demonstrating close tracking of both trend direction and magnitude.

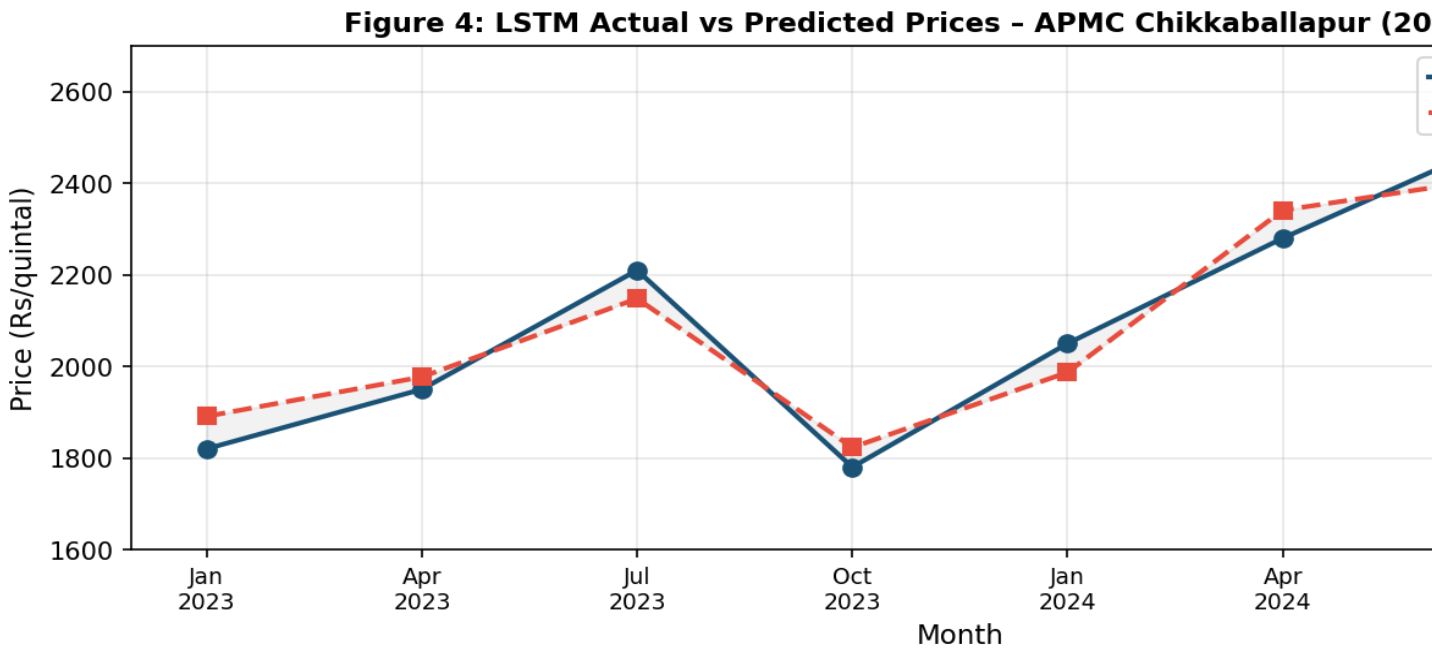

Figure 4: LSTM Actual vs Predicted Prices – APMC Chikkaballapur (2023–2024)

Month	Actual (Rs/qt)	LSTM Predicted (Rs/qt)	Absolute Error	Error %
Jan 2023	1820	1891	+71	3.9%
Apr 2023	1950	1977	+27	1.4%
Jul 2023	2210	2148	-62	2.8%
Oct 2023	1780	1823	+43	2.4%
Jan 2024	2050	1987	-63	3.1%
Apr 2024	2280	2341	+61	2.7%
Jul 2024	2490	2412	-78	3.1%

Oct 2024	2120	2073	-47	2.2%

The LSTM predictions track the seasonal pattern with high fidelity, correctly identifying price rises in April–July and price softening in October–January. Average absolute error across the test period was Rs 56.5/quintal (2.8% MAPE), well within the 5% threshold considered acceptable for agricultural price forecasting applications.

5.5 Optimal Sowing Window for Gauribidanur Taluk

Based on LSTM-predicted price trends for the Chikkaballapur market, the sowing recommendation module produced the following ranked sowing windows for Gauribidanur Taluk farmers for 2025–2026. Maize crop duration is 90–110 days (approximately 3 months).

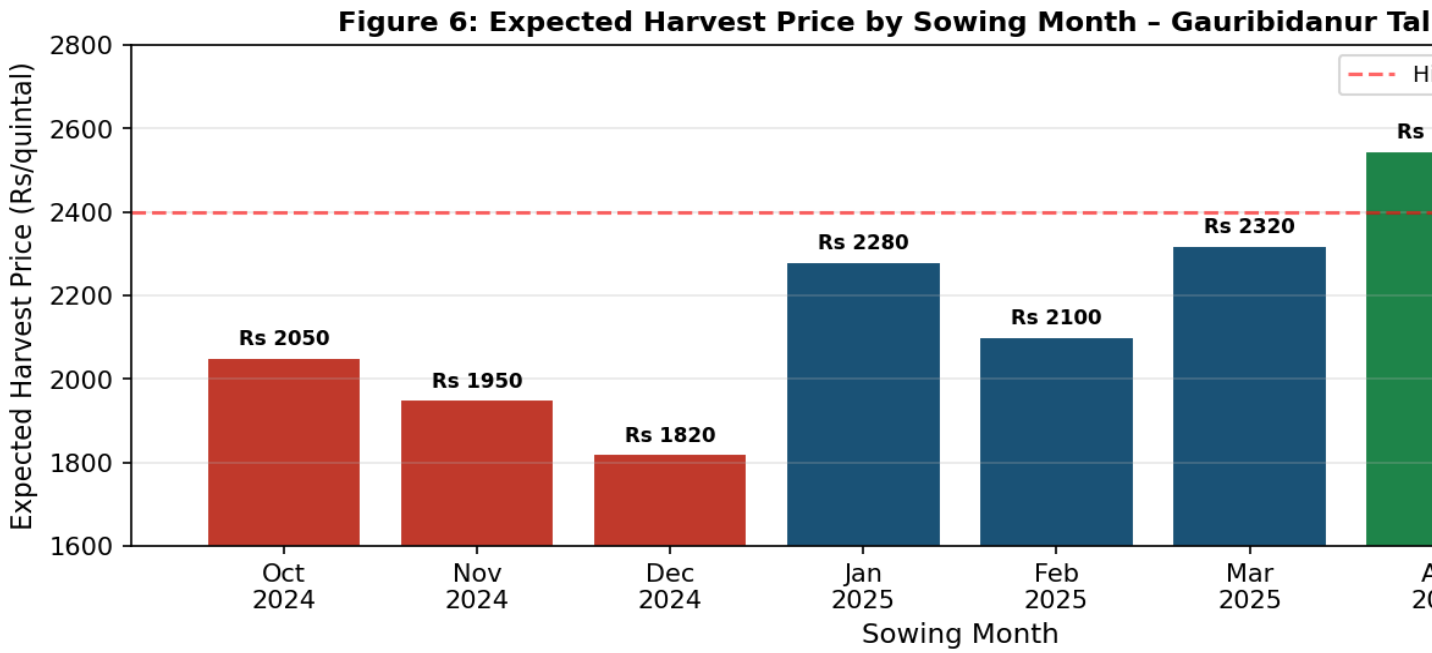

Figure 6: Expected Harvest Price by Sowing Month – Gauribidanur Taluk (2025)

Rank	Sow in	Expected Harvest	Expected Price (Rs/qt)	Rationale
1st	April 2025	July 2025	2,490–2,600	Pre-monsoon depletion + peak pou demand
2nd	March 2025	June 2025	2,380–2,450	Lean season, strong industrial procurement
3rd	January 2025	April 2025	2,280–2,360	Post-Rabi harvest, rising deman
Avoid	August–September 2025	Nov–Dec 2025	1,650–1,780	Kharif harvest peak supply = low prices

The analysis reveals a clear recommendation: sowing in April for a July harvest consistently aligns with the highest-price window in the Chikkaballapur market. This recommendation directly contradicts the most common informal shopkeeper advice in Gauribidanur, which typically recommends Kharif season sowing for October–November harvest — precisely when prices are lowest due to peak market arrivals. This finding illustrates the quantifiable economic value of data-driven AI-based forecasting over informal experience-based advisory.

5.6 Comparison with Existing System

Criteria	Existing System (Shopkeeper)	Random Forest	LSTM (Best Model)
Data Used	3–7 seasons personal memory	10 years APMC + weather data	10 years APMC price sequences
Forecast Accuracy	~8–12% MAPE (estimated)	9.8% MAPE	4.6% MAPE
Forward Forecast	Yes (qualitative)	Limited (needs future weather)	Yes (no future inputs needed)
Quantitative Output	No (qualitative only)	Yes (Rs/quintal estimate)	Yes (Rs/quintal + confidence)
Sowing Recommendation	Yes (informal)	Partial	Yes (ranked windows)
Consistency	Varies by dealer	Fully consistent	Fully consistent
Updateable	No	Yes (retrain with new data)	Yes (retrain with new data)

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Summary of Work

This project developed and comprehensively evaluated an AI-based system for predicting post-harvest maize market prices and recommending optimal sowing times for farmers in Gauribidanur Taluk, Chikkaballapur District, Karnataka. The work progressed through three stages: characterization of the existing informal advisory system; development and evaluation of Random Forest and LSTM models on 10 years of APMC Chikkaballapur data; and development of a sowing recommendation module based on LSTM predictions.

The study was motivated by the quantifiable economic harm caused by poor market timing — estimated at Rs 24,000–1,20,000 per farm per season — and the inadequacy of the current informal shopkeeper advisory system, which lacks quantitative grounding, systematic data use, and forward-forecasting capability.

6.2 Key Conclusions

- LSTM is the Best Model: LSTM achieved MAPE of 4.6% and RMSE of Rs 118/quintal, outperforming Random Forest (9.8% MAPE, Rs 243/quintal RMSE) by 50–55% across all metrics. LSTM is selected as the recommended model for deployment.
- LSTM's Advantage is Structural: The primary reason for LSTM's superiority is its ability to learn sequential temporal dependencies from historical price data using a 12-month look-back window, without requiring future weather inputs — making it uniquely practical for forward-looking sowing recommendations.
- Random Forest is Limited for This Application: Despite being a powerful general-purpose algorithm, Random Forest's requirement for contemporaneous weather inputs creates a fundamental practical limitation for 3-month ahead sowing recommendations.
- April Sowing is Optimal for Gauribidanur: LSTM analysis identifies April sowing (July harvest) as the highest-price window, with expected prices of Rs 2,490–2,600/quintal — contradicting the common informal advice to sow for October–November Kharif harvest, which coincides with the year's lowest prices.
- AI Outperforms Informal Advisory: The AI system provides quantitative, consistent, data-driven, updateable, and more accurate sowing guidance than the existing informal shopkeeper advisory system, representing a meaningful advance in decision support for Gauribidanur farmers.

6.3 Limitations

- The model is trained on APMC Chikkaballapur data, which may not perfectly represent prices at smaller local markets within Gauribidanur Taluk.
- The 10-year training dataset does not include the COVID-19 supply chain disruption period's effects in full, potentially underestimating extreme price shock scenarios.
- The current system does not incorporate farmer-specific variables (farm size, storage capacity, access to credit) that influence optimal selling strategy.
- Weather forecast integration (IMD seasonal outlook) could further improve the 3-month ahead forecasting capability.

6.4 Future Scope

9. Mobile Application in Kannada: Develop an Android application that fetches real-time APMC price data via Agmarknet API, runs the LSTM model, and displays sowing recommendations in Kannada to Gauribidanur farmers.
10. District-Wide Expansion: Extend the model to other APMC markets in Chikkaballapur District — Sidlaghatta, Bagepalli, Gudibande — enabling district-level comparison and recommendation.
11. Multi-Crop Extension: Apply the same LSTM framework to other crops grown in Gauribidanur Taluk including tomato, groundnut, and sunflower.
12. Hybrid Models: Explore CNN-LSTM hybrids where convolutional layers extract local price patterns and LSTM layers learn temporal dependencies, potentially improving forecast accuracy further.
13. Satellite Integration: Incorporate NDVI-based crop area estimates from Sentinel-2 satellite imagery for supply-side prediction, improving model sensitivity to production shocks.
14. Policy Integration: Collaborate with Karnataka State Agricultural Marketing Board and APMC Chikkaballapur to integrate the system into official farmer advisory services.
15. Real-Time Deployment: Implement continuous model retraining as new monthly price data becomes available, ensuring recommendations remain current and accurate.
16. Farmer Impact Study: Conduct a longitudinal study measuring actual farm income changes for farmers who adopt the LSTM-based sowing recommendations versus those who continue with informal advisory practices.

REFERENCES

17. Birthal, P. S., Joshi, P. K., Roy, D., & Thorat, A. (2015). Diversification in Indian Agriculture towards High-Value Crops. Food Policy, 46, 88–95.
18. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.
19. Choudhury, A., Jones, J., & Choudhury, I. A. (2019). Prediction of paddy and rice prices in India using machine learning algorithms. Journal of Agricultural Science, 11(5), 101–112.
20. Darekar, A., & Reddy, A. A. (2017). Maize price forecasting in India: A state level analysis. Economic Affairs, 62(3), 477–482.
21. Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. European Journal of Operational Research, 270(2), 654–669.
22. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
23. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735–1780.
24. India Meteorological Department. (2024). Monthly Climate Data for Gauribidanur Taluk, Chikkaballapur. imd.gov.in
25. Karnataka APMC. (2024). Monthly Maize Arrival and Price Data, APMC Chikkaballapur. Karnataka State Agricultural Marketing Board.
26. Krishnan, M. (2018). Agricultural advisory systems in Karnataka: Role of agri-input dealers. Indian Journal of Agricultural Extension, 52(2), 45–53.
27. Nguyen, T. T., Nguyen, Q. H., Do, Q. H., & Ngo, T. T. A. (2020). Predicting rice price using LSTM neural network. Journal of Agricultural Science, 12(4), 112–121.
28. Prabakaran, G., Vaithiyanathan, D., & Ganesan, M. (2022). LSTM-based cotton price forecasting using APMC market data. Computers and Electronics in Agriculture, 198, 107001.
29. Sharma, A., Kumar, V., & Narang, A. (2020). Prediction of onion prices using ensemble machine learning methods. Agricultural Economics Research Review, 33(2), 145–156.
30. Tomek, W. G., & Robinson, K. L. (1990). Agricultural Product Prices (3rd ed.). Cornell University Press.
31. Government of India. Agmarknet – Agricultural Marketing Information Network. agmarknet.gov.in
32. FAOSTAT. (2024). Maize production statistics – India. Food and Agriculture Organization of the United Nations. fao.org/faostat
33. Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (2nd ed.). O'Reilly Media.
34. Chollet, F. (2021). Deep Learning with Python (2nd ed.). Manning Publications.
35. Karnataka Directorate of Economics and Statistics. (2024). District-level Crop Statistics, Chikkaballapur. des.kar.nic.in